Victor Petrov

650-382-7248 | 799vpetrov@gmail.com | linkedin.com/in/vpetrov-codes | v7petrov.github.io

Education

California Polytechnic University, San Luis Obispo — San Luis Obispo, CA
*Bachelor of Science in Computer Engineering, **GPA 3.7/4.0*** — *June 2026*

Relevant Coursework: Real Time Operating Systems, Classical Controls, Modern Computer Architecture, Data Structures & Algorithms, Digital Signals & Systems, Computer Networking

Experience

Controls Software Engineer Intern — June 2025 – Sep 2025
General Atomics — *Poway, CA*

- Maintained and debugged a large-scale C++ real-time flight simulator codebase (200K+ lines), identifying and resolving critical issues across multiple interconnected systems with hard real-time networking requirements
- Optimized embedded OpenCV and ONNX Runtime implementations through hardware-level profiling and code tuning, achieving 300% faster inference speeds and reducing frame processing latency by 65%
- Led cross-platform migration effort, porting a complex Windows-based flight simulator (150+ source files) to Linux, addressing OS-specific dependencies and system calls while maintaining feature parity
- Reengineered CMake configuration files to reduce build times by 45% and improve cross-platform compilation
- Developed Python and Bash automation scripts to eliminate 15+ hours of manual build and deployment work

Projects

Real-Time Motor Control System | *C/C++, Embedded Linux, Bare-metal Firmware*

- Debugged and adapted open-source motor controller firmware in C for custom PCB hardware, resolving initialization failures, PWM signal generation issues
- Validated error handling routines in firmware, testing robust fault detection for overcurrent, overvoltage, and overtemperature conditions
- Calibrated motor controller parameters based on telemetry data analysis, optimizing performance while maintaining safety margins

GridGuard – Residential Battery UPS | *Li-ion Cell Validation, Power Electronics, BMS Concepts*

- Designed power management circuitry including charge/discharge control, overcurrent protection, and grid-backup switching for a residential power supply
- Led full product development cycle from customer discovery to working prototype, making design tradeoffs across battery chemistry, cost, and safety requirements

Full Stack 4-Axis Robot Arm | *PCB Design, Bare Metal C, MicroPython, I2C, BLE*

- Designed and brought up compact motor controller PCBs using RP2040, enabling distributed control architecture with I2C communication between 4 nodes
- Developed BLE-enabled web application interface for wireless robot control and telemetry monitoring

Autonomous VTOL Simulation | *Python, C++, Gazebo, MAVSDK*

- Implemented hardware-in-the-loop (HIL) testing using Gazebo simulation and MAVSDK, validating autonomous mission logic before physical deployment
- Designed finite state machines for autonomous flight modes and integrated real-time computer vision via OpenCV

Function Generator and Oscilloscope | *C++, STM32, I2C, SPI, UART, JTAG*

- Programmed function generator and oscilloscope on STM32 without hardware abstraction layer (HAL), using direct register-level access and debugging through JTAG
- Implemented I2C, UART, and SPI communication protocols from STM32 reference manuals, demonstrating deep microcontroller architecture knowledge

Technical Skills

Languages: C, C++ (Modern C++17), Python, MATLAB, Bash/Shell
Embedded Systems: Bare-metal Firmware, FreeRTOS, RTOS, Device Drivers, Hardware Bring-up, STM32, RP2040, ESP32
Protocols: I2C, SPI, UART, TCP/UDP, CAN (familiar), AXI
Tools: Oscilloscope, Logic Analyzer, Multimeter, CMake, Git, Docker, JTAG, MATLAB/Simulink, Gazebo